

SUMMER INTERNSHIP PRESENTATION REPORT

C++ Programming Internship

Console Applications, OOP & File Handling in C++

STUDENT CREDENTIALS

Student Name	Raja Yadav
Enrollment / Roll No.	25SCS1003000861
Program & Semester	3rd Semester
Host Organization	Thiranex – Skill Development & Future Tech
Internship Duration	30 Jul 2026 – 29 Aug 2026
Intern ID	THX-JUL3026-503

Candidate & Program Overview

Key details of the student and the internship program

S Candidate Details

Student Name	Raja Yadav
Roll / Enrollment No.	25SCS1003000861
Semester	3rd Semester
University	IILM University, Greater Noida, U.P.
Found Internship Via	Thiranex

P Internship Profile

Role Offered	Intern – C++ Programming
Host Organization	Thiranex – Skill Development & Future Tech
Duration	30 Jul – 29 Aug 2026 (~4.5 Weeks)
Intern ID	THX-JUL3026-503
Modules Completed	4 of 4 (100% on-track)

Internship at a Glance

A quick snapshot of the internship engagement

1

ORGANIZATION

Thiranex

2

DURATION

30 Jul – 29 Aug 2026

3

DOMAIN

C++ Programming

4

MODE

Remote / Online

ABOUT THE ROLE

Undertook a structured, milestone-based C++ Programming internship at Thiranex, covering core problem-solving, object-oriented programming and file handling — from a console-based Student Management System, through a Bank Management Application and Library Management System, to an interactive Mini Game Project. Each of the four assigned modules was submitted on time and marked “Completed Successfully” on the official tracking portal.

Internship Offer Letter

Formal offer extended by Thiranex confirming the internship terms

OFFER DETAILS

Intern Name

Raja Yadav

Intern ID

THX-JUL3026-503

Internship Role

Intern – C++ Programming

Commencement Date

30 Jul 2026

Completion Date

29 Aug 2026

Work Mode

Remote / Project-Based

Issued By

Thiranex – Skill Development & Future Tech



About the Organization

Thiranex — Skill Development & Future Tech

Organization Overview

Thiranex is a skill development and future-technology focused platform offering structured, project-based internship programs in modern web development. The program builds industry-relevant, hands-on competencies through progressively advanced, real-world coding assignments, with continuous progress tracking via an online submission portal.

- ✓ Structured, milestone-based internship tracks
- ✓ Emphasis on modern, industry-standard tools & practices
- ✓ Task-driven learning with defined deliverables and deadlines
- ✓ Continuous progress tracking via an online submission portal

CREDENTIALS

Program Type

Skill Development Internship

Focus Areas

C++ Programming, Future Tech

Tracking

Online Milestone Portal

Certification

Issued on Successful Completion

Objectives of the Internship

What the internship set out to achieve

1

Master Core C++ Concepts

Apply structures, classes and object-oriented programming principles to real-world style problems.

2

Implement File Handling

Use file I/O to persist records reliably across program runs.

3

Build Menu-Driven Applications

Design clean, navigable console interfaces for multi-feature applications.

4

Apply Programming Logic

Use loops, arrays and conditional logic to implement interactive game mechanics.

Module 1: Student Management System

Due Date: 06 Aug 2026 • Status: Completed Successfully

1

TASK DESCRIPTION

Developed a console-based Student Management System in C++ that efficiently manages student records using file handling and menu-driven operations.

KEY FEATURES IMPLEMENTED

- ✓ Add, update, delete, and display student information
- ✓ Used file handling to store data persistently
- ✓ Implemented menu-driven operations for smooth navigation
- ✓ Validated user input across all CRUD operations

EXPECTED OUTCOME

A functional application that allows users to add, update, delete, and display student information with persistent data storage for reliable record management.

✓ Completed Successfully

Module 2: Bank Management Application

Due Date: 13 Aug 2026 • Status: Completed Successfully

2

TASK DESCRIPTION

Designed and implemented a C++ based Bank Management Application that simulates core banking operations using object-oriented programming and file handling.

KEY FEATURES IMPLEMENTED

- ✓ Included features like deposit, withdrawal, and balance check
- ✓ Used object-oriented programming concepts (classes & objects)
- ✓ Stored customer data securely using file management
- ✓ Implemented input validation for all banking transactions

EXPECTED OUTCOME

A secure and functional banking system capable of performing deposits, withdrawals, and balance inquiries while maintaining persistent customer records.

✓ Completed Successfully

Module 3: Library Management System

Due Date: 20 Aug 2026 • Status: Completed Successfully

3

TASK DESCRIPTION

Developed a Library Management System that efficiently manages books, members, and borrowing records using structured or object-oriented programming.

KEY FEATURES IMPLEMENTED

- ✓ Implemented structures/classes for book and member details
- ✓ Supported book issue and return features
- ✓ Included search functionality by title or author
- ✓ Persisted library records using file handling

EXPECTED OUTCOME

A functional application that enables book addition, issue and return processing, and search by title or author for streamlined library operations.

✓ Completed Successfully

Module 4: Mini Game Project (Tic Tac Toe / Snake Game)

Due Date: 27 Aug 2026 • Status: Completed Successfully

4

TASK DESCRIPTION

Created a console-based mini game in C++ that demonstrates core programming concepts such as loops, arrays, and conditional logic.

KEY FEATURES IMPLEMENTED

- ✓ Used loops, arrays, and condition statements for game logic
- ✓ Displayed the game board dynamically after each move
- ✓ Added win/loss detection and replay options
- ✓ Structured game logic for clean, readable code

EXPECTED OUTCOME

An interactive game with dynamic board display, win/loss detection, and replay functionality, showcasing effective implementation of game logic.

✓ Completed Successfully

Tools & Technologies Used

The stack applied across all four modules

1

C++ Language

Core syntax, control structures, functions

2

Object-Oriented Programming

Classes, objects, encapsulation

3

File Handling

fstream-based persistent data storage

4

Structures & Arrays

Structured records and data collections

5

STL Concepts

Vectors, strings, and standard containers

6

Compiler / IDE

GCC / Code::Blocks / Dev-C++

Learning Outcomes & Skills Gained

Technical and professional takeaways from the internship

Core C++ Programming

- ✓ Structures, classes & object-oriented design
- ✓ Control flow, loops & arrays
- ✓ Functions & modular code organization

Data Persistence

- ✓ File handling with fstream (read/write)
- ✓ Structuring persistent records
- ✓ Data validation & error handling

Application Design

- ✓ Menu-driven console interfaces
- ✓ CRUD-style record management
- ✓ Game logic & interactive programs

Challenges & Key Learnings

Reflections from the internship journey



Managing file I/O reliably

Learned to handle file-open failures and data corruption gracefully, ensuring records persist correctly across program runs.



Structuring OOP designs

Moved from procedural, function-heavy code to well-encapsulated classes and objects for the Bank Management Application.



Designing clean menu systems

Refined input validation and loop-driven menus so invalid choices no longer crashed or confused the console applications.



Implementing game logic

Learned to manage 2D array-based game state and win/loss conditions cleanly for the Tic Tac Toe / Snake mini game.

Certificate of Completion

Official recognition of successful internship completion



Certificate ID: THX-JUL3026-503 | Issued by Thiranex – Skill Development & Future Tech | Signed: Hariharan M, Founder & CEO

Conclusion

This internship at Thiranex provided a comprehensive, hands-on progression through core C++ programming — from structured record-keeping systems to an interactive console-based game. Each milestone built directly on the last, resulting in a strong, practical foundation in object-oriented programming, file handling and application design, along with the confidence to independently design and build C++ console applications.

4/4

Milestones Completed

100%

On-Track Submissions

4

C++ Applications Built

Thank You

Raja Yadav | 3rd Semester | IILM University, Greater Noida